

Vidath Dassanayake

COMPUTER SCIENCE UNDERGRADUATE | SOFTWARE ENGINEERING

Colombo, Sri Lanka • +94 71 147 9009 • dassanayakevidath@gmail.com • vidathdassanayake.me • github.com/VidathVi
• linkedin.com/in/vidathdassanayake

PROFESSIONAL SUMMARY

DevOps and software engineer building cloud-native systems on GCP with Kubernetes, Terraform, and CI/CD. Experienced in OOP, container optimization, and infrastructure as code (IaC), and working in Agile teams to ship production-style projects.

TECHNICAL SKILLS

Languages:	Java, Python, TypeScript, JavaScript, C# (Unity), HTML, CSS, OOP
Frameworks & Libraries:	Next.js, React, Node.js, Express.js, Prisma ORM, PyTorch
Databases:	PostgreSQL, Redis, SQL
Cloud & DevOps:	Docker, Kubernetes, GKE, GCP, Terraform, Helm, GitHub Actions, Kong
Other:	Postman, Unity 6, Blender, Git, Linux, Bash

PROJECTS

OreanStudio (Orean 360 Platform)

2025 – 2026

Technologies: Next.js, React, Node.js 20, Express.js, TypeScript, Zod, PostgreSQL, Prisma ORM, BullMQ, Redis, FFmpeg, Kubernetes, GKE, Terraform, Helm, Kong API Gateway, GitHub Actions

- Built a Next.js 16 frontend and a Node.js 20 / Express 5 REST API in TypeScript (Prisma ORM, JWT auth, Zod validation) on a 14-model PostgreSQL schema.
- Implemented six OAuth integrations (Google, YouTube, Meta, LinkedIn, TikTok, WhatsApp), with BullMQ workers for scheduled publishing (retries with exponential backoff, publish audit logs) and FFmpeg video transcoding into five platform presets.
- Implemented KEDA-driven autoscaling for BullMQ workers, scaling from 0 to 10 pods on Redis queue depth.
- Cut the frontend Docker image from 1.5 GB to ~150 MB (90%) using a three-stage node:20-alpine build with Next.js standalone output.
- Automated Prisma migrations and Helm deployments to GKE through GitHub Actions CI/CD, with atomic rollback.
- Collaborated with a 6-member engineering team using Agile/Scrum practices to design and deliver the platform.

GEO Engine (Zero-Cost LLM Pipeline)

2025 – 2026

Technologies: Python, PyTorch, Unsloth, Qwen2.5, LoRA, vLLM, Firecrawl, Playwright, Docker Compose, WSL2, Modal

- Built a full-stack GEO (Generative Engine Optimisation) application that optimises content for LLMs like ChatGPT and Gemini.
- Built a Python data-collection pipeline with self-hosted Firecrawl in Docker, round-robin IP rotation across three free VPN regions (via WSL2), and a Playwright session manager for LinkedIn with 10–25 second request delays; gathered 377 programming posts from 15 subreddits (score ≥ 100 , upvote ratio $\geq 85\%$) through the Arctic Shift archive.
- Cleaned the scraped data (minimum-length filter, deduplication, spam and all-caps removal, PII stripping) down to 374 records and formatted them as ChatML, split 299/37/38 into train, validation and test sets.
- Fine-tuned Qwen2.5-7B-Instruct with LoRA (Unsloth, 4-bit) on a free Google Colab T4 GPU; training loss fell from 1.84 to 0.96 over 3 epochs at \$0 compute cost.
- Served the merged model with vLLM on Modal (A10G GPU, scale-to-zero after 120 seconds idle) behind an OpenAI-compatible API used by the Orean backend, within Modal's free credit tier.

Modly Blender Extension

2025

Technologies: Python, Blender Python API (bpy), FastAPI

- Developed a Blender 5.2 add-on integrating Modly's local AI 3D generation via a custom Node Graph, with a dynamic node tree generated from JSON manifests and an isolated FastAPI backend to prevent GPU conflicts.

CERTIFICATIONS

IBM SkillsBuild: [Cloud Computing Fundamentals](#)

Cisco: [Python Essentials 2](#)

Boot.dev: [Docker](#), [Git](#), [Linux](#), [SQL](#), [Python](#), [OOP in Python](#)

DataCamp: Java, Java OOP

LANGUAGES

English:	Fluent
Sinhala:	Native

EDUCATION

Informatics Institute of Technology (IIT) – BSc (Hons) Computer Science · Year 3 of 4 · Expected graduation: 2028

Modules – Object-Oriented Programming (Java/Python), Databases, Client-Server Architecture, Web Development

Kingswood College – A/Ls: Applied Mathematics & Pure Mathematics (2021 – 2023)